

Adaptive Alpha Strategy Backtest

Methodology, Cell-by-Cell Explanation, Assumptions, Variables, and CAPM Alpha Interpretation

Prepared from the notebook code supplied in this conversation

1. Executive summary

Assignment coverage. The notebook does address the assignment structure: it defines a universe and strategy, performs an event-driven historical backtest, evaluates return/volatility/Sharpe/max drawdown on a 252-trading-day rolling window against buy-and-hold, and estimates CAPM alpha and beta using excess returns.

Important accuracy point. The current Alpha 43 implementation contains a unit inconsistency: ADV20 is calculated from dollar volume (Close x Volume), while the Alpha 43 numerator is share volume. The canonical Alpha 43 construction uses volume / ADV20 where ADV20 is average share volume. If the strategy is described as using canonical Alpha 43, Cell 5 should be corrected before final submission.

Numerical alpha. A numerical CAPM alpha cannot be truthfully inferred from source code alone. The actual alpha depends on the downloaded prices and the executed backtest. No regression output values were supplied here. A separate notebook cell is provided with this report that generates a PDF from the live notebook variables and therefore prints the exact full-sample and subperiod alpha, beta, p-values, confidence interval, and conclusion after the notebook has been run.

2. Required correction before claiming canonical Alpha 43

Replace the current ADV20 calculation in Cell 5 with average share volume:

```
ADV20_PANEL = VOLUME_PANEL.rolling(window=ADV_WINDOW, min_periods=ADV_WINDOW).mean()
```

Reason: the current code computes $VOLUME_PANEL / \text{average}(\text{Close} \times \text{Volume})$, which mixes share volume with dollar volume and has units of 1/price. That is not the intended relative-volume ratio. `DOLLAR_VOLUME_PANEL` may still be retained for diagnostics, but it should not be used as Alpha 43 ADV20 if you are following the WorldQuant-style formula.

Naming note. Your Alpha 42 and Alpha 43 scores are further cross-sectionally re-ranked and centered around zero before being used as directional signals. That is a defensible strategy transformation, but it means the trading score is better described as an “Alpha 42-derived score” and “Alpha 43-derived score,” rather than the untouched canonical alpha value.

3. Cell-by-cell explanation

Cell	What it does	Main output	Key assumptions / caveats
1	Imports	Loads pandas, NumPy, yfinance, statsmodels and Plotly.	Libraries are available; Yahoo Finance is acceptable for an academic backtest; Plotly is used for visualisation.
2	Backtest parameters	Defines asset, market, risk-free proxy, universe, dates, signal windows, rolling window, and costs.	NVDA is the traded asset; SPY proxies the market; ^IRX proxies risk-free; 252 trading days/year; fixed current-stock universe; no point-in-time membership; zero transaction and borrow costs initially.

3	Historical downloads	Downloads OHLCV for the signal universe, SPY and ^IRX.	Yahoo data quality is sufficient; auto-adjusted stock prices are used; end date is exclusive.
4	Cleaning and alignment	Extracts OHLCV panels, aligns dates, builds main df, forward-fills Treasury yield.	Selected stock and SPY trading dates define the calendar; missing peer observations are allowed; last available Treasury observation is carried forward.
5	Returns and inputs	Creates asset/market open-to-open returns, daily risk-free rate, VWAP proxy, dollar volume and ADV20.	Holding interval is open-to-open; ^IRX is treated approximately as an annual percentage yield; typical price is a VWAP proxy; risk-free information is lagged to avoid look-ahead. ADV20 needs the correction described above for Alpha 43.
6	Signal/regime engine	Computes Alpha 10, Alpha 101, momentum confirmation, trend measures, efficiency ratio, state-machine regimes, Alpha 42/43-derived mean-reversion signals and Desired_Long.	Long-only; fixed thresholds; cross-sectional ranks use the chosen universe; 3-of-4 persistence for regime detection; 5-day smoothing; 20/60-day trends; ER thresholds are researcher-chosen; no parameter re-estimation.
7	Execution/backtest	Converts Desired_Long into actual trades at next-day open, applies breakeven exit rule, cash/risk-free accrual, costs, mark-to-market returns and wealth.	Fractional shares; full capital invested on entry; no shorting; exit signal from prior close; if exit would realize a loss, hold until breakeven; intraday high touching limit is assumed fillable; no slippage/market impact; cash earns risk-free when flat for full interval.
Diagnostic block	Regime diagnostics	Prints regime counts, percentages, and Signal_Raw by regime.	Signal_Raw is desired position, not necessarily actual Position because breakeven rule may override exit.
7A	Price/regime/trade plot	Plots price, VWAP proxy, regime markers, actual buys and exits.	Visual diagnostic only; close is plotted while execution is generally at open/limit price.
7B	Actual position plot	Plots long/cash state and days where exit is pending.	Position is end-of-execution-day state; exit-pending means strategy wanted out but breakeven rule prevented sale.
8	CAPM excess returns	Computes strategy and market returns above risk-free.	Same daily risk-free series is subtracted from strategy and SPY open-to-open returns.
9	Performance function	Computes cumulative return, CAGR-style annualized return, annualized volatility, Sharpe and max drawdown.	252-day annualization; arithmetic Sharpe scaling; daily observations are treated as the return frequency.
10	Overall comparison	Compares strategy, asset buy-and-hold and SPY on summary performance statistics.	Buy-and-hold uses the same effective backtest period as strategy after warm-up.
11	Growth of \$1	Plots cumulative wealth for strategy, selected stock and SPY.	No external cash flows; reinvestment is implicit through return compounding.
12	Rolling 1-year performance	Calculates rolling 252-day return, volatility, Sharpe and maximum drawdown for strategy and asset buy-and-hold.	252 trading days approximates one year; every window overlaps; this is rolling performance evaluation, not walk-forward parameter re-fitting.
13	Rolling Sharpe plot	Plots rolling 1-year Sharpe for strategy vs asset buy-and-hold.	Same risk-free series and 252-day annualization are used in both.
14	Rolling return plot	Plots rolling 1-year compounded return for strategy vs buy-and-hold.	Each point represents the trailing 252-trading-day return.
15	Rolling volatility plot	Plots rolling annualized volatility.	Daily standard deviation is scaled by sqrt(252).
16	Rolling max drawdown plot	Plots worst peak-to-trough loss within each trailing 252-day window.	Each rolling window begins at wealth 1.0 so an immediate first-day loss is counted.
16A	Full-period drawdown	Plots historical drawdown path from the running peak across the entire backtest.	This is different from rolling max drawdown; it is the actual drawdown path from inception.
17	CAPM function	Runs OLS of strategy excess returns on market excess returns with intercept, HAC standard errors,	SPY is market proxy; intercept is daily alpha; annual alpha = daily alpha x 252; HAC maxlags=5 is a

		annualized alpha, beta, p-values, t-statistic, R-squared and CI.	modelling choice; linear CAPM is assumed.
18	Full-sample CAPM + plot	Runs the full-sample CAPM and plots observations plus fitted line.	Daily observations; regression line is descriptive of linear market exposure.
19	Regression output	Prints the statsmodels regression summary.	HAC robust inference changes standard errors/p-values, not fitted coefficients.
20	Subperiod robustness	Splits sample into earlier period and most recent two years and reruns CAPM.	Two-year split is based on END_DATE rather than actual last trading date; this tests stability but is not a rolling CAPM.

4. Main variables and parameters

Variable	Current value	Meaning
ASSET_TICKER	NVDA	Single stock actually traded.
MARKET_TICKER	SPY	CAPM market proxy and benchmark.
RF_TICKER	^IRX	13-week Treasury bill yield proxy.
START_DATE	2019-01-01	Download start date.
END_DATE	2026-09-05	Exclusive yfinance end date.
SIGNAL_UNIVERSE	25 large/liquid US stocks + selected asset	Cross-sectional ranking universe; not point-in-time.
ALPHA10_WINDOW	4	Window used by Alpha 10 persistence logic.
ALPHA43_PRICE_LOOKBACK	7	Close change lookback in Alpha 43.
ADV_WINDOW	20	ADV20 window.
ROLLING_WINDOW	252	Trailing performance window.
TRADING_DAYS	252	Annualization constant.
TRANSACTION_COST_BPS	0.0	Per-unit position-change cost; zero in current setup.
SHORT_BORROW_BPS_ANNUAL	0.0	Unused in long-only strategy.
ALPHA_CONFIRM_WINDOW	5	Smoothing horizon for momentum confirmation.
MOMENTUM_SHORT_WINDOW	20	Short trend horizon.
MOMENTUM_LONG_WINDOW	60	Long trend horizon.
EFFICIENCY_WINDOW	20	Efficiency-ratio lookback.
MOMENTUM_ER_ENTRY	0.30	Minimum ER to enter momentum regime.
MOMENTUM_ER_EXIT	0.20	ER below which momentum regime exits.
MEAN_REVERSION_ER_THRESHOLD	0.20	ER below which non-momentum state may be mean-reverting.
BREAKEVEN_INCLUDE_TRADING_COSTS	True	Raises breakeven threshold enough to recover entry/exit costs.
BREAKEVEN_BUFFER_BPS	0.0	Extra required gain above breakeven.

4A. Key derived variables / DataFrame columns

Variable / column	Meaning
ASSET_OPEN/HIGH/LOW/CLOSE/VOLUME columns	Adjusted daily OHLCV for the traded stock.
ASSET_RETURN_COL	Selected-stock open-to-open return: $\text{Open}_t / \text{Open}_{(t-1)} - 1$.
MARKET_RETURN_COL	SPY open-to-open return on the same dates.
RF_Yield	Forward-filled ^IRX close quote.
RF_Annual	$\text{RF_Yield} / 100$.
RF_Daily_Raw	Approximate effective daily risk-free return from RF_Annual.
Rf_Daily	Lagged daily risk-free return aligned to the strategy holding interval.
ASSET_VWAP_COL	Typical-price proxy: $(\text{High} + \text{Low} + \text{Close}) / 3$.
ALPHA10_COL	Cross-sectional Alpha 10 rank for selected stock.
ALPHA101_COL	Intraday momentum ratio $(\text{Close} - \text{Open}) / (\text{High} - \text{Low} + 0.001)$.
Alpha_Momentum_Confirmation	50% smoothed Alpha-10 persistence + 50% smoothed Alpha-101 strength.
Trend_20D_Return	20-trading-day close-to-close trend return.
Trend_60D_Return	60-trading-day close-to-close trend return.
Efficiency_Ratio	Absolute net 20-day move divided by sum of absolute daily moves.
Momentum_Direction	State-machine regime direction: +1 up, -1 down, 0 no established

	momentum.
ALPHA42_COL	Centered cross-sectional Alpha 42-derived mean-reversion score.
ALPHA43_COL	Centered cross-sectional Alpha 43-derived mean-reversion score.
Regime	Upward Momentum, Downward Momentum, Mean Reversion or Neutral.
Desired_Long	Signal intent: 1 wants long; 0 wants cash/exit; NaN before signals are ready.
Signal_Raw	Compatibility alias of Desired_Long.
Position	Actual executed state: 1 long, 0 cash.
Entry_Price	Original open execution price of current long position.
Breakeven_Price	Minimum permitted exit price under the no-realized-loss rule.
Exit_Pending	True when exit is desired but blocked because price is below breakeven.
Trade_Action	Human-readable execution action for the date.
Held_Regime	Reason/current state of the holding, including Breakeven Hold.
Exit_Price	Actual exit execution price when a trade closes.
Realized_Trade_Return	Price return from original entry to actual exit, before portfolio-level interpretation.
Unrealized_Return	Open-price mark-to-market return versus entry while position remains open.
Trade	1 when a buy or exit occurs, 0 otherwise.
Trading_Cost	Transaction-cost impact divided by start-of-interval wealth.
Cash_Weight	1 - Position.
Strategy_Return	Event-driven portfolio return for the open-to-open interval.
Strategy_Wealth	Cumulative product of 1 + Strategy_Return.
Strategy_Excess	Strategy_Return - Rf_Daily.
Market_Excess	SPY open-to-open return - Rf_Daily.
rolling_results	DataFrame of trailing 252-day strategy and buy-and-hold return, volatility, Sharpe and max drawdown.

5. Complete assumption set

Data and universe

- Yahoo Finance is accurate enough for the exercise.
- The chosen stock universe is fixed using today's surviving/liquid names; survivorship and selection bias can remain.
- Corporate actions are handled through yfinance auto-adjusted prices; adjusted prices are treated as investable price series for academic purposes.
- Peer stocks may have missing observations; cross-sectional ranks ignore missing names.

Signal construction

- Signals are computed from information available by the close and are acted on at the next open.
- Typical price $(High + Low + Close)/3$ is only a proxy for true VWAP.
- Cross-sectional ranks are meaningful only relative to the selected universe.
- Momentum regime thresholds, smoothing windows and ER cutoffs are fixed ex ante in the code; they are not learned or re-estimated over time.
- Alpha 42/43 are transformed into centered directional scores; this is a strategy design choice, not the untouched published formula.

Execution

- Long-only: Position is 0 or 1; short signals are ignored.
- The strategy can buy fractional shares and deploy all available capital.
- Orders generated from a closing signal execute at the next day's open.
- When an exit would realize a loss, the strategy keeps holding until the original breakeven threshold is reached.
- If an exit is pending and intraday High reaches the breakeven limit, the code assumes the limit order fills exactly at breakeven. This can be optimistic because a printed high does not guarantee executable queue priority or volume.
- No slippage, bid-ask spread, market impact, taxes, financing margin, liquidity limits, or partial fills are modelled.

- Transaction cost is currently 0 bps, so current headline results are frictionless unless changed.
- When flat for the full open-to-open interval, cash earns the daily risk-free rate.

Risk-free rate

- $\wedge$ IRX is treated as an annual percentage rate by dividing by 100 and converting to a 252-day effective daily rate.
- This is an approximation because Treasury bill quotes follow market conventions that are not exactly an effective annual yield.
- The risk-free series is lagged so the rate used for a holding interval is information available before entry.

Performance

- 252 trading days approximates one year.
- Rolling windows overlap substantially; adjacent rolling statistics are therefore highly dependent.
- Rolling analysis evaluates realized strategy performance; it does not re-fit parameters or perform walk-forward optimization.
- Buy-and-hold comparisons use the same post-warm-up effective sample as the strategy.

CAPM and inference

- SPY is used as the market factor, not the full investable market portfolio.
- CAPM is assumed linear in daily excess returns.
- The regression intercept is interpreted as daily abnormal return; annualized alpha is intercept x 252.
- HAC/Newey-West style covariance with maxlags=5 is used to reduce sensitivity to heteroskedasticity and short-horizon autocorrelation.
- The 5-lag choice is not uniquely correct and should be disclosed.
- Statistical significance is evaluated using the alpha p-value and confidence interval; a positive point estimate alone is not sufficient evidence of skill.

6. How to interpret the CAPM alpha correctly

The regression being estimated is: **Strategy excess return = alpha + beta x Market excess return + error**

Alpha (intercept). The daily intercept measures the strategy return not explained by contemporaneous SPY excess returns under the one-factor CAPM. The code reports annualized alpha as daily alpha x 252.

Beta. Beta measures how strongly the strategy co-moves with the SPY excess return. Beta near 1 means equity-market-like exposure; beta below 1 means lower sensitivity; negative beta would imply inverse sensitivity.

Statistical test. For the assignment question “is alpha significant?”, the key evidence is the alpha p-value and 95% confidence interval. A conventional conclusion is significant at the 5% level when $p < 0.05$ and the confidence interval excludes zero.

Economic significance. Even statistically significant alpha should be compared with transaction costs, drawdown, turnover, sample length, and robustness across subperiods. A large backtest alpha that disappears in the recent two-year subperiod is weak evidence of persistent skill.

7. Conclusion logic to use after running the notebook

Case	Condition	Correct wording
Positive and significant	Annualized alpha > 0 and p-value < 0.05	Evidence is consistent with positive CAPM alpha in-sample. Then verify recent-period alpha, costs, and robustness before claiming persistent skill.
Positive but not significant	Annualized alpha > 0 and p-value ≥ 0.05	The backtest has a positive alpha estimate,

		but the evidence is insufficient to distinguish it reliably from zero.
Negative and significant	Annualized alpha < 0 and p-value < 0.05	The strategy underperforms the CAPM benchmark on a risk-adjusted basis in a statistically significant way.
Negative but not significant	Annualized alpha < 0 and p-value >= 0.05	The estimate is negative, but statistical evidence is insufficient to conclude persistent negative alpha.

Current numerical conclusion: Not determinable from the code text alone because the fitted regression output was not provided. Do not invent the alpha. Run the notebook through Cell 20, then run the supplied PDF-generator cell. It will insert the exact alpha, beta, t-statistic, p-value, R-squared, confidence interval, rolling statistics, and subperiod conclusions into the final PDF.

8. Recommended wording for the assignment

A concise methodology statement you can use:

“We backtest a long-only adaptive strategy on the selected stock using signals formed at the close and executed at the next market open. Strategy performance is evaluated against buy-and-hold using overlapping 252-trading-day rolling windows for compounded return, annualized volatility, Sharpe ratio and maximum drawdown. We then regress daily strategy excess returns on SPY excess returns using an intercept and HAC-robust standard errors. The intercept is interpreted as CAPM alpha and is assessed using its t-statistic, p-value and 95% confidence interval. We also repeat the CAPM regression over an earlier sample and the most recent two years to assess robustness.”

9. Final accuracy checklist before submission

- Fix ADV20 for Alpha 43 to use average share volume if claiming canonical Alpha 43.
- Run all cells in order after the fix; do not mix outputs from old and new runs.
- Confirm Cell 16 is the rolling 1-year maximum drawdown plot and keep Cell 16A as full-history drawdown.
- Record the exact full-sample annualized alpha and alpha p-value from Cell 17/18.
- Compare full-sample alpha with the recent two-year alpha from Cell 20.
- Disclose zero transaction costs unless you rerun with a non-zero assumption.
- Do not describe the breakeven exit rule as eliminating losses; the portfolio remains marked to market while underwater.
- Describe the cross-sectional universe and survivorship-bias limitation.
- Do not call the rolling evaluation a walk-forward parameter optimization; parameters are fixed.